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7

8

A

B

C

D

E

F

power_charge

File: power_charge.kicad_sch

battery_supply

File: battery_supply.kicad_sch

mcu

File: mcu.kicad_sch

sensing

File: sensing.kicad_sch

force

File: force.kicad_sch

stage_position

File: stage_position.kicad_sch

actuator_x

File: actuator_x.kicad_sch

actuator_y

File: actuator_y.kicad_sch

storage_comm

File: storage_comm.kicad_sch

user_interface

File: user_interface.kicad_sch

fault_handling

File: fault_handling.kicad_sch

Active stabilisation pen – research electronics Rev A. PROPOSED DESIGN: not built, not measured.
Every part marked VERIFY or SELECT in its fields must be checked against the named source before layout.
Power tree: USB–C –> MCP73831 –> cell (BQ29700 + 2 NMOS) –> TLV75530 (+3V0, +3V0A) and TPS22917 (VM0T, gated).
Calculations: electronics/calcs/drive_sense.py –> results/electronics/; ngspice drive stage: electronics/spice/.

RESEARCH ELECTRONICS – DEVELOPMENT DRAFT – NOT RELEASED FOR FABRICATION

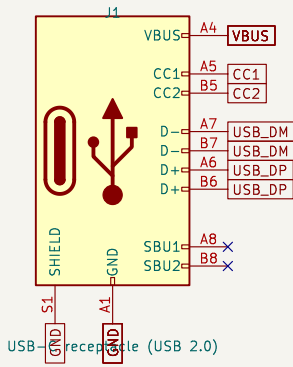
Active stabilisation pen programme

Sheet: /
File: pen_research.kicad_sch

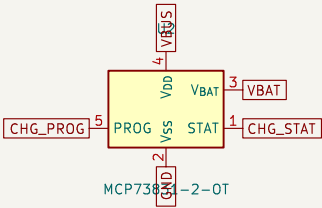
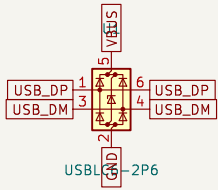
Title: Active stabilisation pen – research electronics Rev A

Size: A3Date: 2026–09–27Rev: A

KiCad E.D.A. 8.0.9Id: 1/12



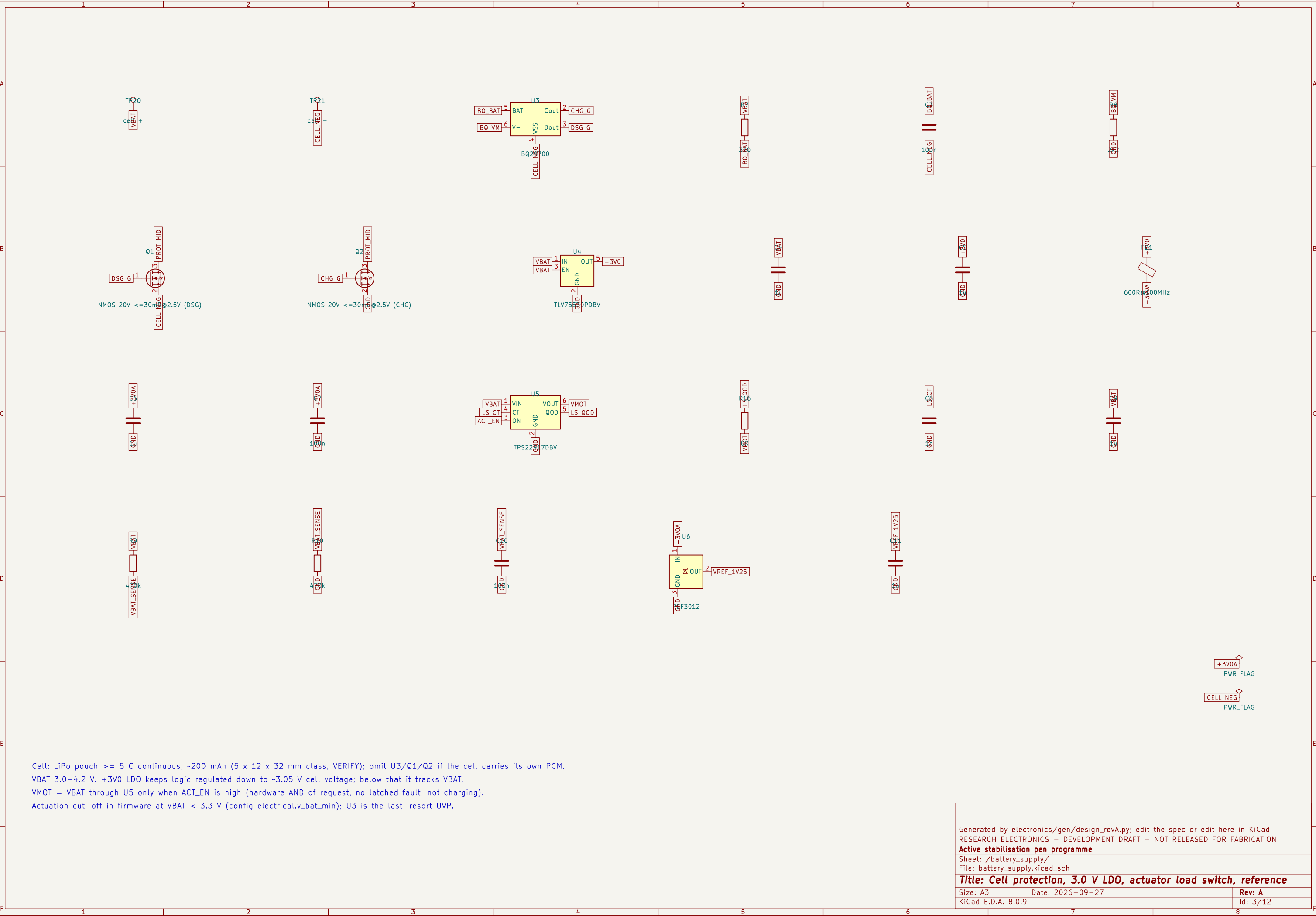
USB-C receptacle (USB 2.0)



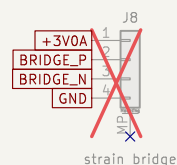
GND
PWR_FLAG

VBUS
PWR_FLAG

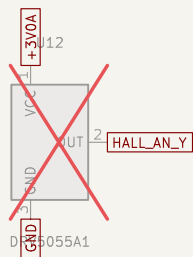
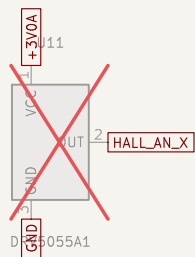
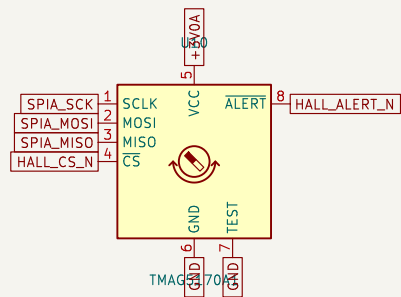
CHG_DET = VBUS/2 (2.37–2.62 V for 4.75–5.25 V VBUS): valid logic high at VDD 3.0 V, below VDD+0.3 V.
CHG_DET also drives the charging interlock (sheet fault_handling): actuation is disabled while VBUS is present.
Charge current 147 mA (R3): check against the selected cell's 0.5–1 C limit and the 43 C skin limit.







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TMA65170 at the lever tail reads a sense magnet on the carrier: Bx, By → stage angles; Bz → axial slide s → $F_{ax} = F_{pre} + k_{ax} s$.
Cross-axis and actuator-field coupling calibrated on the bench (EXP-B04); sample mid-PWM to avoid coil-field ripple.

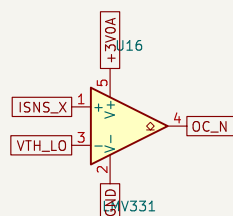
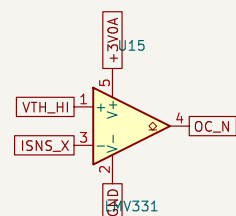
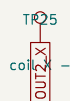
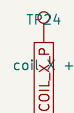
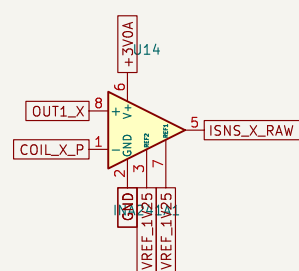
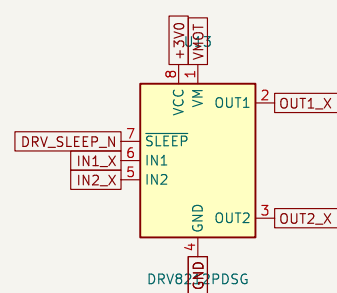
Generated by electronics/gen/design_revA.py; edit the spec or edit here in KiCad
RESEARCH ELECTRONICS – DEVELOPMENT DRAFT – NOT RELEASED FOR FABRICATION

Active stabilisation pen programme

Sheet: /stage_position/
File: stage_position.kicad_sch

Title: Stage angle and axial slide sensing (3-axis Hall)

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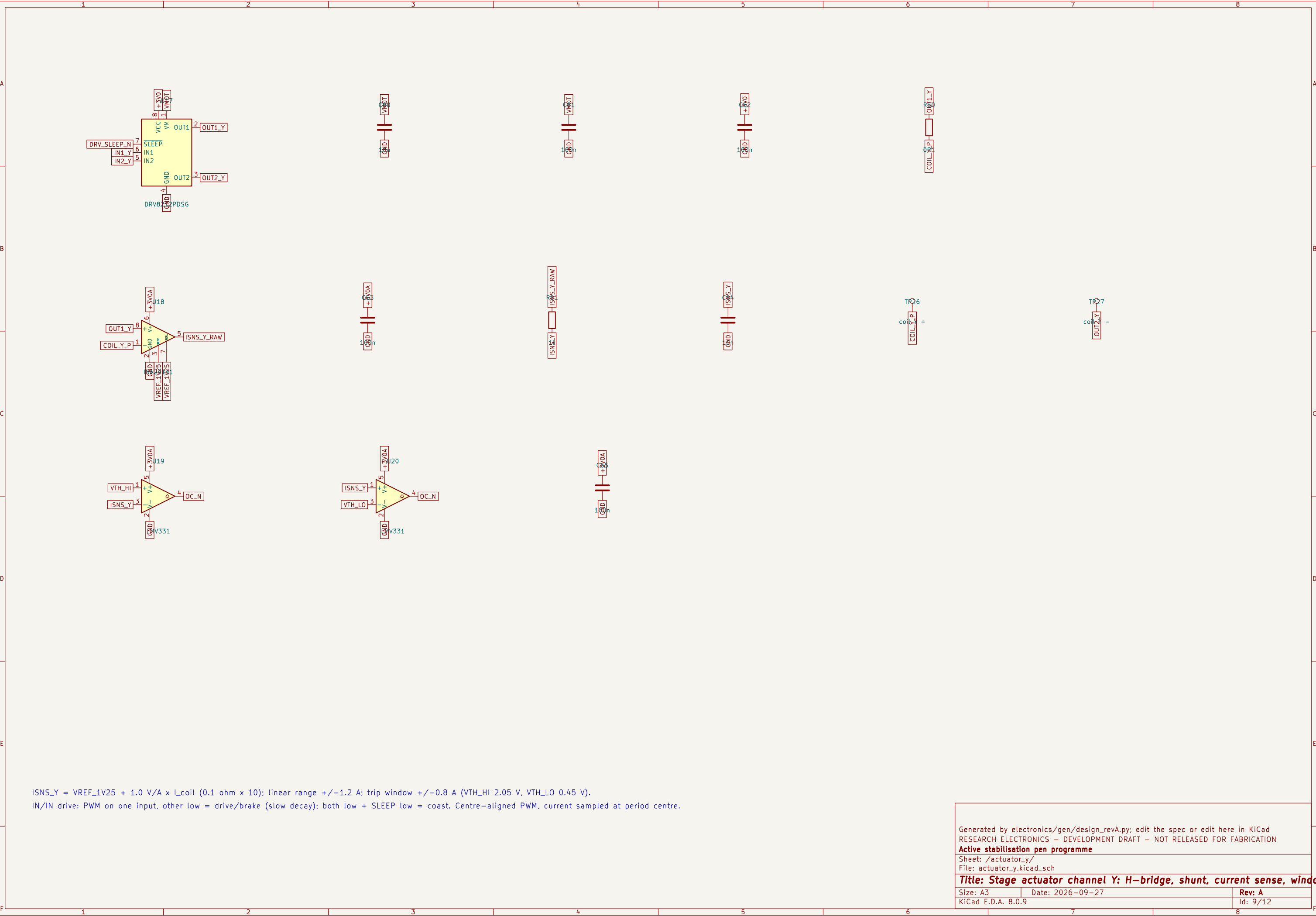
ISNS_X = VREF_1V25 + 1.0 V/A x I_coil (0.1 ohm x 10); linear range +/-1.2 A; trip window +/-0.8 A (VTH_HI 2.05 V, VTH_LO 0.45 V).
IN/IN drive: PWM on one input, other low = drive/brake (slow decay); both low + SLEEP low = coast. Centre-aligned PWM, current sampled at period centre.

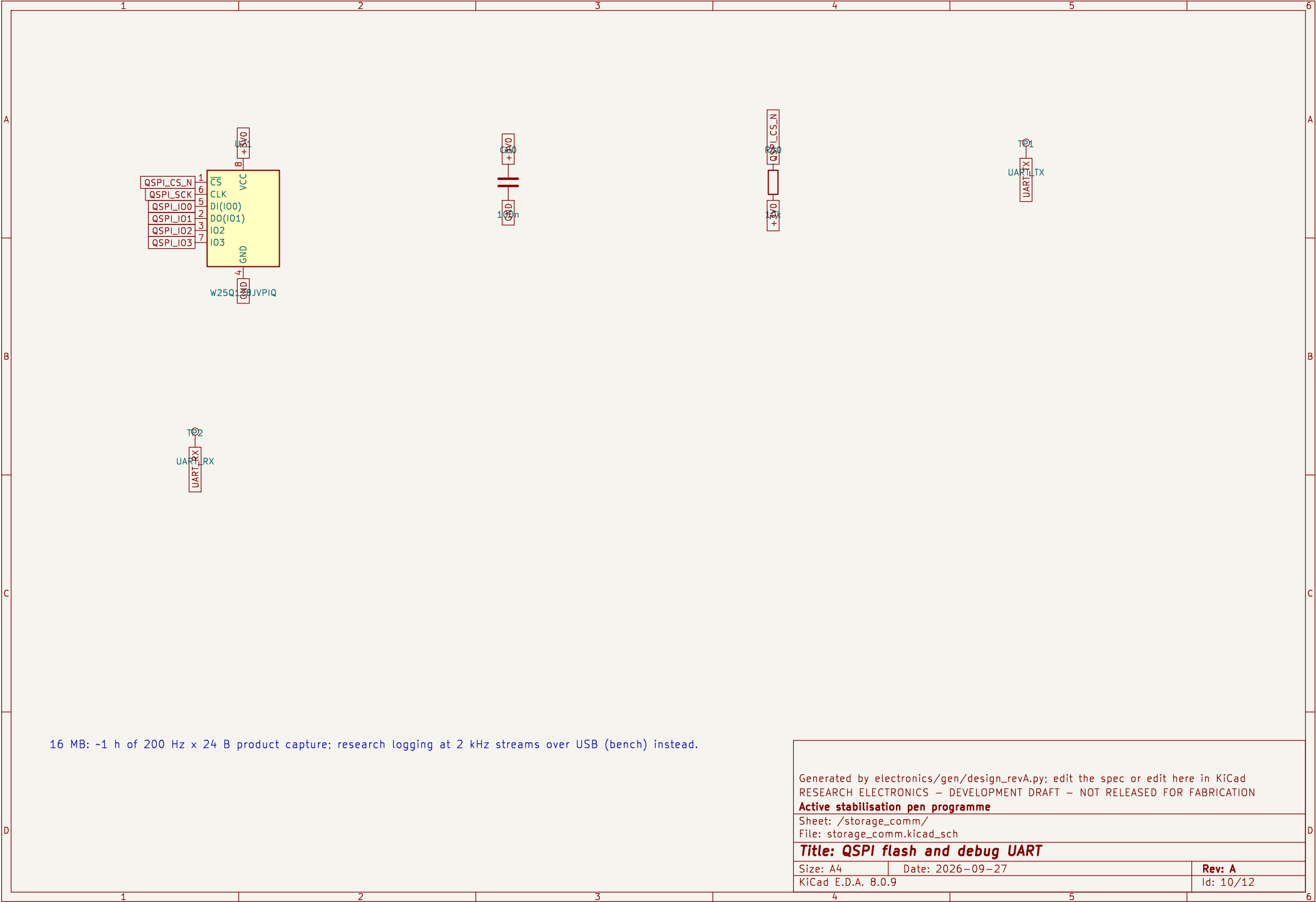
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RESEARCH ELECTRONICS – DEVELOPMENT DRAFT – NOT RELEASED FOR FABRICATION
Active stabilisation pen programme

Sheet: /actuator_x/
File: actuator_x.kicad_sch

Title: Stage actuator channel X: H-bridge, shunt, current sense, window com

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16 MB: ~1 h of 200 Hz x 24 B product capture; research logging at 2 kHz streams over USB (bench) instead.

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Active stabilisation pen programme

Sheet: /storage_comm/
File: storage_comm.kicad_sch

Title: QSPI flash and debug UART

Size: A4	Date: 2026-09-27
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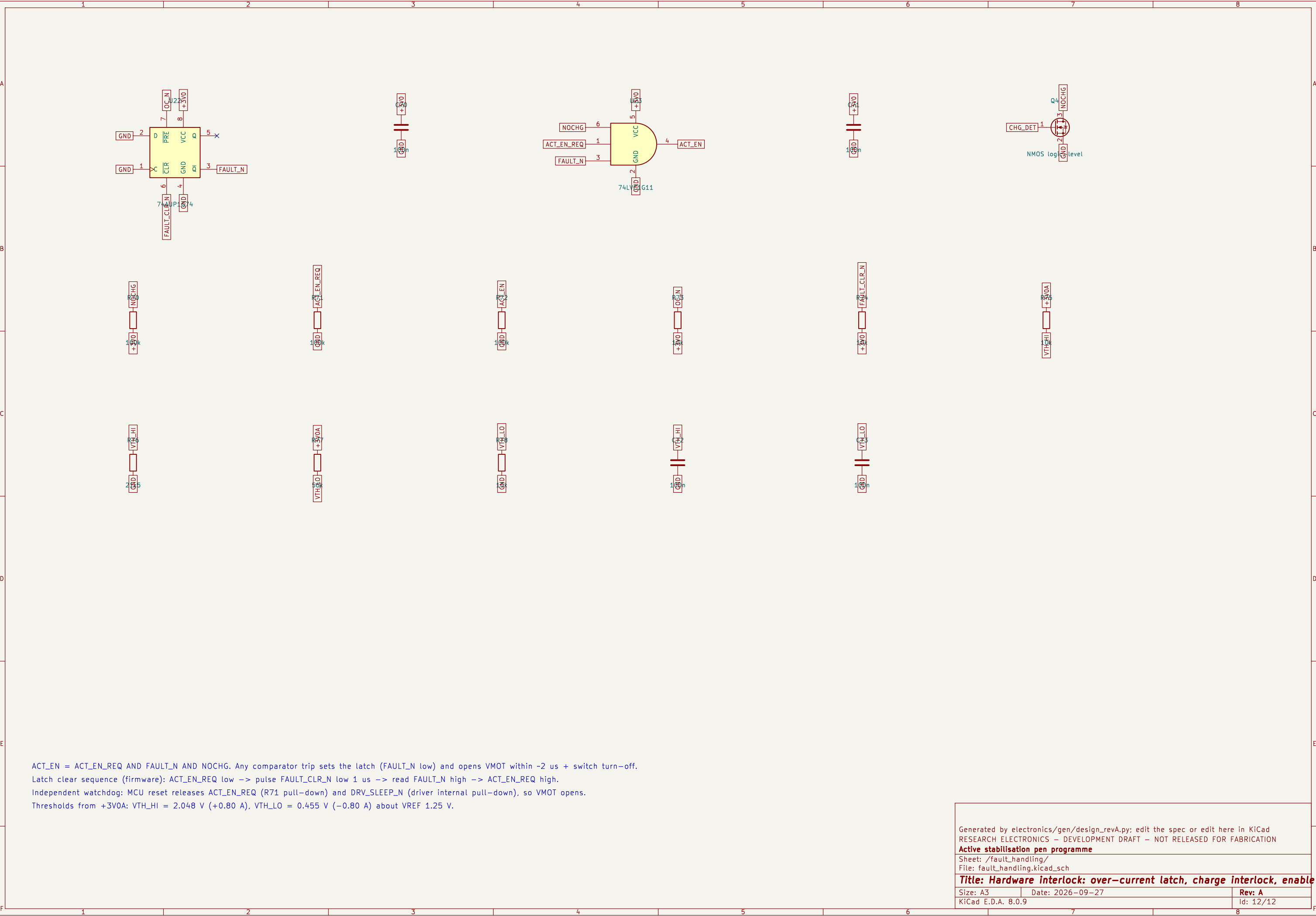
Rev: A

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Id: 10/12



ERM: coin vibration motor on flying leads (TP28/TP29); limit PWM duty to its rated voltage.



ACT_EN = ACT_EN_REQ AND FAULT_N AND NOCHG. Any comparator trip sets the latch (FAULT_N low) and opens VMOT within ~2 us + switch turn-off.
Latch clear sequence (firmware): ACT_EN_REQ low -> pulse FAULT_CLR_N low 1 us -> read FAULT_N high -> ACT_EN_REQ high.
Independent watchdog: MCU reset releases ACT_EN_REQ (R71 pull-down) and DRV_SLEEP_N (driver internal pull-down), so VMOT opens.
Thresholds from +3V0A: VTH_HI = 2.048 V (+0.80 A), VTH_LO = 0.455 V (-0.80 A) about VREF 1.25 V.